Advanced Material Physics

Duy M. Dao

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Part I

Dielectric Polarization under Static Electric Field

Unit system used in this note. Throughout this note, the unit system is chosen such that $4\pi\epsilon_0 = 1$.

1 Multipole Expansion of a Point Cloud

Potential of a point cloud and the need for an expansion. Consider N point charges q_i at positions $\vec{r}_i$, which together form a point cloud. The electric potential at $\vec{R}$ is the sum of the contributions of all point charges,

$$\phi(\vec{R}) = \sum_{i=1}^{N} \frac{q_i}{|\vec{R} - \vec{r}_i|}.$$

In general, this potential depends on the charge positions in a complicated way. Far from the cloud, however, the multipole expansion breaks the potential down into multipole terms. From now on, $R \equiv |\vec{R}|$ and $r_i \equiv |\vec{r}_i|$ denote the distances from the origin, and $\hat{R} = \vec{R}/R$ denotes the unit vector along $\vec{R}$.

Expansion of the inverse distance in powers of r_i/R. Factoring R out of the distance gives

$$\frac{1}{|\vec{R} - \vec{r}_i|} = \frac{1}{R}\left(1 - 2\,\frac{\hat{R}\cdot\vec{r}_i}{R} + \frac{r_i^2}{R^2}\right)^{-1/2} = \frac{1}{R}\,(1+\eta)^{-1/2}, \qquad \eta = -2\,\frac{\hat{R}\cdot\vec{r}_i}{R} + \frac{r_i^2}{R^2}.$$

Far from the cloud, $r_i \ll R$ for every charge, so $|\eta| \ll 1$ and the Taylor expansion of $(1+\eta)^{-1/2}$ applies,

$$\begin{aligned}(1+\eta)^{-1/2} &= 1 - \frac{1}{2}\eta + \frac{3}{8}\eta^2 + \mathcal{O}(\eta^3) \\ &= 1 - \frac{1}{2}\left(-2\,\frac{\hat{R}\cdot\vec{r}_i}{R} + \frac{r_i^2}{R^2}\right) + \frac{3}{8}\left[4\,\frac{(\hat{R}\cdot\vec{r}_i)^2}{R^2} - 4\,\frac{(\hat{R}\cdot\vec{r}_i)\,r_i^2}{R^3} + \frac{r_i^4}{R^4}\right] + \mathcal{O}(\eta^3).\end{aligned}$$

Grouping the terms by order of r_i/R and dropping the third and higher orders gives

$$(1+\eta)^{-1/2} = 1 + \frac{\hat{R}\cdot\vec{r}_i}{R} + \left[\frac{3}{2}\,\frac{(\hat{R}\cdot\vec{r}_i)^2}{R^2} - \frac{1}{2}\,\frac{r_i^2}{R^2}\right] + \mathcal{O}\left(\frac{r_i^3}{R^3}\right).$$

Multipole expansion of the potential. Substituting this series into the potential gives, to second order in r_i/R,

$$\boxed{\phi(\vec{R}) \approx \sum_{i=1}^{N} \frac{q_i}{R}\left\{1 + \frac{\hat{R}\cdot\vec{r}_i}{R} + \left[\frac{3}{2}\,\frac{(\hat{R}\cdot\vec{r}_i)^2}{R^2} - \frac{1}{2}\,\frac{r_i^2}{R^2}\right]\right\}.} \tag{1.1}$$

The three terms in braces are the monopole, dipole, and quadrupole terms.

Monopole term: the total charge. The first term is the monopole term,

$$\boxed{\sum_{i=1}^{N} \frac{q_i}{R} = \frac{Q}{R}, \qquad Q = \sum_{i=1}^{N} q_i,} \tag{1.2}$$

where Q is the total charge, also called the monopole moment, a scalar or rank-0 tensor. This term vanishes when the point cloud is neutral.

Dipole term: the dipole moment vector. The second term is the dipole term,

$$\boxed{\sum_{i=1}^{N} \frac{q_i\,(\hat{R}\cdot\vec{r}_i)}{R^2} = \frac{\hat{R}\cdot\vec{p}}{R^2}, \qquad \vec{p} = \sum_{i=1}^{N} q_i\,\vec{r}_i,} \tag{1.3}$$

where $\vec{p}$, a vector or rank-1 tensor, is the dipole moment vector. This term remains even when the point cloud is neutral.

Quadrupole term written as a matrix product. The third term is the quadrupole term,

$$\sum_{i=1}^{N} \frac{q_i}{R}\left[\frac{3}{2}\frac{(\hat{R}\cdot\vec{r}_i)^2}{R^2} - \frac{1}{2}\frac{r_i^2}{R^2}\right] = \frac{1}{2R^3}\sum_{i=1}^{N} q_i\left[3(\hat{R}\cdot\vec{r}_i)^2 - r_i^2\right].$$

From now on, Greek indices $\alpha, \beta \in \{x, y, z\}$ label Cartesian components, and the Latin index i still labels the charges. For example, $r_{i\alpha}$ is the α component of $\vec{r}_i$. In components, the squared dot product becomes a double sum,

$$(\hat{R}\cdot\vec{r}_i)^2 = \sum_{\alpha}\hat{R}_\alpha r_{i\alpha}\sum_{\beta}\hat{R}_\beta r_{i\beta} = \sum_{\alpha,\beta}\hat{R}_\alpha\hat{R}_\beta\, r_{i\alpha}r_{i\beta}.$$

Because $\hat{R}$ is a unit vector, $\sum_{\alpha,\beta}\hat{R}_\alpha\hat{R}_\beta\,\delta_{\alpha\beta} = 1$, where $\delta_{\alpha\beta}$ is the Kronecker delta. The squared distance therefore takes the same form,

$$r_i^2 = \sum_{\alpha,\beta}\hat{R}_\alpha\hat{R}_\beta\, r_i^2\,\delta_{\alpha\beta}.$$

Both sums resemble the inner product $\vec{u}^T M\vec{u}$ of a vector $\vec{u}$ and a matrix M. To see this, let $\vec{w} = M\vec{u}$. Each component $w_\alpha = \sum_\beta M_{\alpha\beta}u_\beta$ multiplies the α-th row of M entry by entry with $\vec{u}$ and sums over the columns β. The inner product is then

$$\vec{u}^T M\vec{u} = \vec{u}^T\vec{w} = \sum_{\alpha} u_\alpha w_\alpha = \sum_{\alpha,\beta} u_\alpha M_{\alpha\beta}u_\beta.$$

Quadrupole moment tensor. Combining the two sums with $u_\alpha = \hat{R}_\alpha$ gives

$$\sum_{i=1}^{N} q_i \left[3(\hat{R}\cdot\vec{r}_i)^2 - r_i^2\right] = \sum_{\alpha,\beta} \hat{R}_\alpha \hat{R}_\beta \sum_{i=1}^{N} q_i \left(3\, r_{i\alpha} r_{i\beta} - r_i^2\, \delta_{\alpha\beta}\right) = \hat{R}^T \mathcal{Q}\, \hat{R}.$$

The quadrupole term is therefore

$$\boxed{\begin{gathered}\sum_{i=1}^{N} \frac{q_i}{R}\left[\frac{3}{2}\frac{(\hat{R}\cdot\vec{r}_i)^2}{R^2} - \frac{1}{2}\frac{r_i^2}{R^2}\right] = \frac{\hat{R}^T \mathcal{Q}\, \hat{R}}{2R^3}, \\ \mathcal{Q}_{\alpha\beta} = \sum_{i=1}^{N} q_i \left(3\, r_{i\alpha} r_{i\beta} - r_i^2\, \delta_{\alpha\beta}\right),\end{gathered}} \tag{1.4}$$

where $\mathcal{Q}$, a symmetric matrix or rank-2 tensor, is the quadrupole moment tensor. The trace of $\mathcal{Q}$ vanishes, because $\sum_\alpha \left(3\, r_{i\alpha}^2 - r_i^2\right) = 0$ for every charge.

Dependence of the moments on the choice of origin. The multipole moments depend on where the origin is placed. Shifting the origin by $\vec{d}$ replaces $\vec{r}_i$ with $\vec{r}_i - \vec{d}$, so the dipole moment becomes

$$\vec{p}\,' = \sum_{i=1}^{N} q_i \left(\vec{r}_i - \vec{d}\right) = \vec{p} - Q\,\vec{d}.$$

The dipole moment is therefore independent of the origin only when $Q = 0$. The same argument shows that $\mathcal{Q}$ is independent of the origin only when Q and $\vec{p}$ both vanish. In general, only the lowest nonvanishing moment is independent of the origin.

2 Polarization

Why the dipole moment describes a neutral material in a field. Most materials are electrically neutral. In an external electric field, the total charge remains zero, but the electrons and the nuclei are displaced in opposite directions and produce an induced dipole. Some materials, such as polar molecular liquids and ferroelectrics, also carry permanent dipoles even without an external field. In both cases, the monopole term vanishes, so the dipole term is the leading term of the multipole expansion far from the material. Because $Q = 0$, the dipole moment is also independent of the choice of origin, as shown in Section 1. The dipole moment is therefore the quantity of interest whenever a material interacts with an electric field.

Polarization as a dipole moment density, and why it is not a derivative. The polarization vector $\vec{P}$ describes the dipole moment density of a material. Real materials consist of discrete dipoles, so the common definition $\vec{P} = \Delta\vec{p}/\Delta V$ cannot be a derivative, because the ratio does not converge as the volume $\Delta V \to 0$. A small volume between atoms contains no dipole and gives zero, whereas a small volume around a point dipole gives a ratio that grows without bound. The ratio is meaningful only for a volume that is large compared with the interatomic distance a but small compared with the length L over which the macroscopic fields vary.

Polarization as a spatial average of the microscopic dipoles. The precise definition replaces the limit with an average. A set of point dipoles $\vec{p}_k$ at positions $\vec{r}_k$ has the microscopic dipole density

$$\vec{P}_{\text{micro}}(\vec{r}) = \sum_k \vec{p}_k\, \delta(\vec{r} - \vec{r}_k),$$

which is a distribution rather than a smooth function. Averaging it with a smooth weighting function w, normalized as $\int w(\vec{r})\, d^3r = 1$ and with a width ℓ that satisfies $a \ll \ell \ll L$, gives the macroscopic polarization

$$\boxed{\vec{P}(\vec{r}) = \int w(\vec{r} - \vec{r}\,')\, \vec{P}_{\text{micro}}(\vec{r}\,')\, d^3r' = \sum_k \vec{p}_k\, w(\vec{r} - \vec{r}_k).} \tag{2.1}$$

This $\vec{P}$ is a density in the integral sense. Over a macroscopic region Ω, the integral $\int_\Omega w(\vec{r} - \vec{r}_k)\, d^3r$ is close to one for a dipole deep inside Ω and close to zero for a dipole far outside it. Only dipoles within a distance ℓ of the boundary contribute partially. The dipole moment of the region is therefore

$$\boxed{\vec{p}_\Omega = \int_\Omega \vec{P}(\vec{r})\, d^3r.} \tag{2.2}$$

Figure 1 illustrates this averaging for a one-dimensional chain of dipoles.

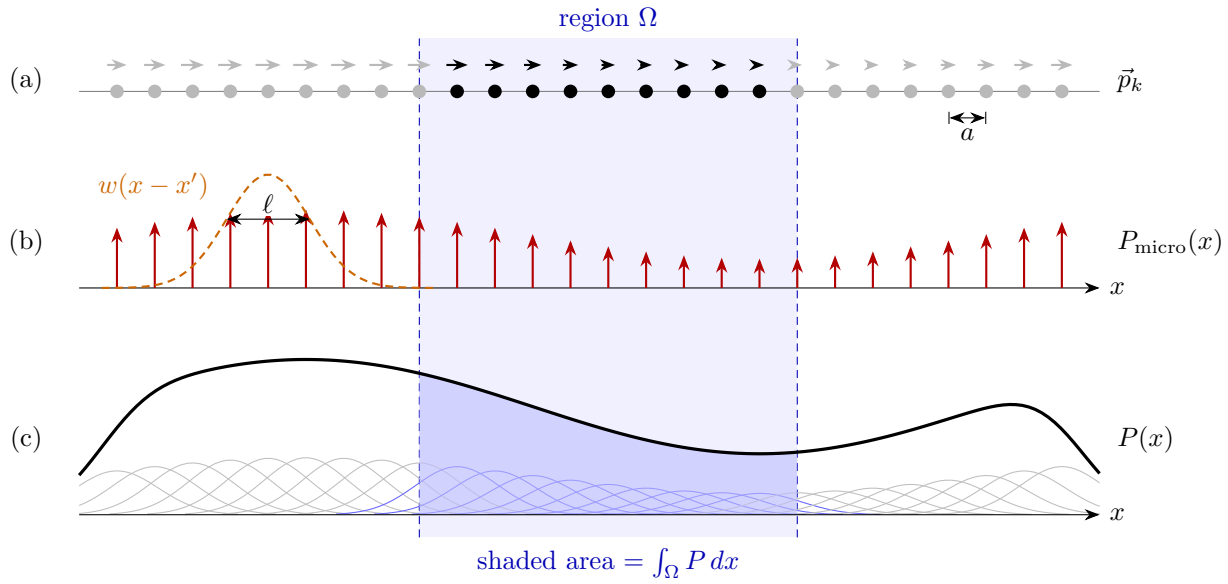


Figure 1: Averaging of a one-dimensional chain of dipoles. (a) Discrete dipoles $\vec{p}_k$ with interatomic distance a, with the dipoles inside Ω in black. (b) Microscopic dipole density as delta peaks, with one averaging window w of width $\ell \gg a$. (c) Contributions $p_k\, w(x - x_k)$ of single dipoles (thin curves) and their sum, the smooth polarization $P(x)$ (thick curve). Dipoles far outside Ω (gray) do not contribute to the shaded area, and dipoles within ℓ of the boundary contribute partially.

Limitation of the averaging picture in periodic crystals. The averaging assumes that the charge can be grouped into localized dipoles $\vec{p}_k$. In a periodic crystal, the electron density is continuous, so the dipole moment of a unit cell depends on how the cell is chosen. Different choices give bulk polarizations that differ by integer multiples of a polarization quantum, which is proportional to a lattice vector divided by the cell volume. Only changes in $\vec{P}$, for example under strain or atomic displacement, are physical. The modern theory of polarization computes these changes from the Berry phase of the Bloch wave functions (King-Smith and Vanderbilt, 1993; Resta, 1994).

Electric field of a single dipole. The field of a volume of dipoles is built from the field of a single dipole. For a point dipole $\vec{p}$ at the origin, the dipole term of the multipole expansion gives the potential

$$\phi(\vec{R}) = \frac{\hat{R}\cdot\vec{p}}{R^2} = \frac{\vec{p}\cdot\vec{R}}{R^3}.$$

The electric field is the negative gradient of the potential, $\vec{E} = -\vec{\nabla}\phi$. With $\vec{\nabla}(\vec{p}\cdot\vec{R}) = \vec{p}$ and $\vec{\nabla}R^{-3} = -3\vec{R}/R^5$,

$$\vec{E}(\vec{R}) = -\frac{\vec{p}}{R^3} + \frac{3(\vec{p}\cdot\vec{R})\,\vec{R}}{R^5}.$$

The field of a point dipole is therefore

$$\boxed{\vec{E}(\vec{R}) = \frac{3(\vec{p}\cdot\hat{R})\,\hat{R} - \vec{p}}{R^3}.} \tag{2.3}$$

Potential of a volume of dipoles. Because $\vec{P}$ is a smooth density, the volume element d^3r at the source point $\vec{r}$ carries the dipole moment $\vec{P}(\vec{r})\,d^3r$. A dipole at $\vec{r}$ instead of the origin has the potential $\vec{p}\cdot(\vec{R}-\vec{r})/|\vec{R}-\vec{r}|^3$. Summing the contributions of all volume elements of a material occupying Ω gives

$$\phi(\vec{R}) = \int_\Omega \frac{\vec{P}(\vec{r})\cdot(\vec{R}-\vec{r})}{|\vec{R}-\vec{r}|^3}\,d^3r.$$

From now on, $\vec{\nabla}_{\vec{r}}$ denotes the gradient with respect to the source point $\vec{r}$, whereas $\vec{\nabla}$ without a subscript still acts on the field point $\vec{R}$. The integrand simplifies with the identity

$$\vec{\nabla}_{\vec{r}}\frac{1}{|\vec{R}-\vec{r}|} = \frac{\vec{R}-\vec{r}}{|\vec{R}-\vec{r}|^3},$$

so that

$$\phi(\vec{R}) = \int_\Omega \vec{P}\cdot\vec{\nabla}_{\vec{r}}\frac{1}{|\vec{R}-\vec{r}|}\,d^3r.$$

Bound charges as the sources of the polarization potential. The product rule $\vec{\nabla}_{\vec{r}}\cdot(f\vec{P}) = f\,\vec{\nabla}_{\vec{r}}\cdot\vec{P} + \vec{P}\cdot\vec{\nabla}_{\vec{r}}f$ with $f = 1/|\vec{R}-\vec{r}|$ splits the integrand into two parts,

$$\phi(\vec{R}) = \int_\Omega \vec{\nabla}_{\vec{r}}\cdot\left(\frac{\vec{P}}{|\vec{R}-\vec{r}|}\right)d^3r - \int_\Omega \frac{\vec{\nabla}_{\vec{r}}\cdot\vec{P}}{|\vec{R}-\vec{r}|}\,d^3r.$$

The divergence theorem turns the first integral into an integral over the boundary $\partial\Omega$, with outward unit normal $\hat{n}$ and surface element dS. The potential of the polarized material is therefore

$$\boxed{\begin{gathered}\phi(\vec{R}) = \oint_{\partial\Omega}\frac{\sigma_b(\vec{r})}{|\vec{R}-\vec{r}|}\,dS + \int_\Omega \frac{\rho_b(\vec{r})}{|\vec{R}-\vec{r}|}\,d^3r, \\ \sigma_b = \vec{P}\cdot\hat{n}, \qquad \rho_b = -\vec{\nabla}_{\vec{r}}\cdot\vec{P},\end{gathered}} \tag{2.4}$$

where σ_b is the bound surface charge density and ρ_b is the bound volume charge density. A polarized material thus produces the same potential as these two charge distributions. In particular, a uniform polarization has $\rho_b = 0$ and produces only a surface charge. Figure 2 shows the origin of both bound charges in a chain of dipoles.

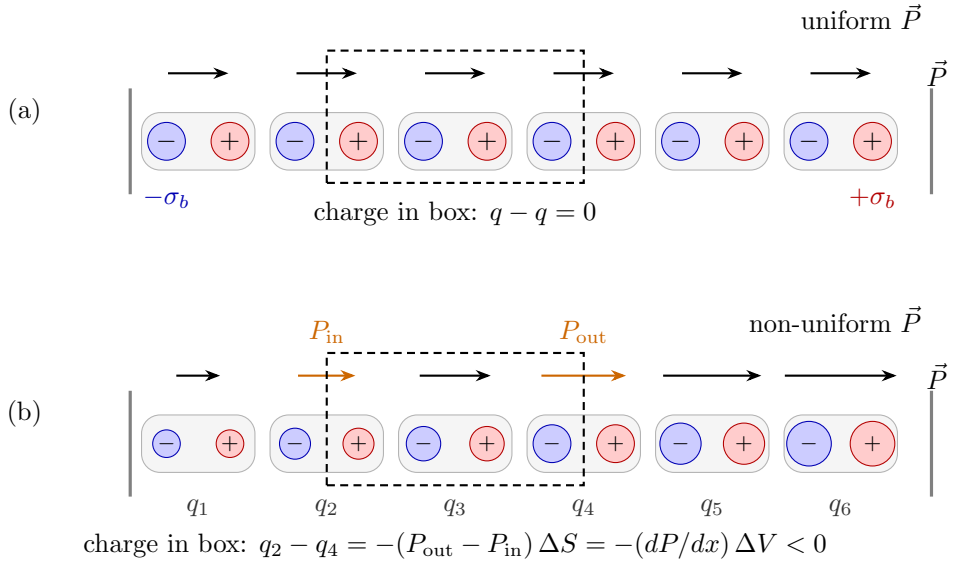


Figure 2: Bound charges of a chain of dipoles between two surfaces. (a) Uniform polarization: the charges of neighboring dipoles cancel inside the box, and only the surfaces carry the uncompensated charges $\pm\sigma_b$. (b) Polarization increasing to the right: the box holds $+q_2$ and $-q_4$ with $q_4 > q_2$, so it carries the net charge $-(dP/dx)\,\Delta V$. Here ΔS is the cross-sectional area of the box and ΔV is its volume.

3 Electric Fields in Dielectric Media

3.1 Gauss's Law in a Polarizable Material

Electric displacement as the field of the free charges. Gauss's law counts both the free charge ρ_f and the bound charge ρ_b,

$$\vec{\nabla} \cdot \vec{E} = 4\pi(\rho_f + \rho_b), \qquad \rho_b = -\vec{\nabla} \cdot \vec{P}.$$

The bound charge is unknown until $\vec{E}$ is known. Moving it to the left side defines the electric displacement $\vec{D}$,

$$\boxed{\vec{D} = \vec{E} + 4\pi\vec{P}, \qquad \vec{\nabla} \cdot \vec{D} = 4\pi\rho_f.} \tag{3.1}$$

This step is only a change of variables. A relation between $\vec{P}$ and $\vec{E}$ is still needed.

Linear response and the dielectric tensor. For a weak field, the polarization is expanded to first order about $\vec{E} = 0$,

$$P_\alpha(\vec{E}) = P_\alpha(0) + \sum_\beta \chi_{\alpha\beta} E_\beta + \mathcal{O}(E^2), \qquad \chi_{\alpha\beta} = \left.\frac{\partial P_\alpha}{\partial E_\beta}\right|_{\vec{E}=0}.$$

With $P_\alpha(0) = 0$ and without the $\mathcal{O}(E^2)$ terms,

$$\boxed{P_\alpha = \sum_\beta \chi_{\alpha\beta} E_\beta, \qquad D_\alpha = \sum_\beta \varepsilon_{\alpha\beta} E_\beta, \qquad \varepsilon_{\alpha\beta} = \delta_{\alpha\beta} + 4\pi\chi_{\alpha\beta},} \tag{3.2}$$

where $\chi_{\alpha\beta}$ is the susceptibility tensor and $\varepsilon_{\alpha\beta}$ is the dielectric tensor. Here $\vec{E}$ is the macroscopic field at the same point, not the external field.

Isotropic materials and the dielectric constant. An isotropic material responds equally in every direction, $\chi_{\alpha\beta} = \chi\,\delta_{\alpha\beta}$, so that

$$\boxed{\vec{P} = \chi\,\vec{E}, \qquad \vec{D} = \varepsilon\,\vec{E}, \qquad \varepsilon = 1 + 4\pi\chi,} \tag{3.3}$$

where χ is the electric susceptibility and ε is the dielectric constant.

Limits of the linear model. Each assumption of the linear model fails in some materials. A ferroelectric carries a spontaneous polarization, $\vec{P}(0) = \vec{P}_s \neq 0$. A strong field, as in nonlinear

optics, makes the second-order term significant,

$$P_\alpha = \sum_\beta \chi_{\alpha\beta} E_\beta + \sum_{\beta,\gamma} \chi^{(2)}_{\alpha\beta\gamma} E_\beta E_\gamma + \mathcal{O}(E^3).$$

A time-dependent field gives a frequency-dependent response, $\vec{P}(\omega) = \chi(\omega)\,\vec{E}(\omega)$. For example, metals at optical frequencies have $\varepsilon(\omega) < 0$. This note treats only static, weak fields in linear isotropic media.

3.2 Boundary Conditions between Two Dielectric Media

Laplace's equation inside each medium. Consider two linear isotropic media, labeled 1 and 2, with constant dielectric constants ε_1 and ε_2 and $\rho_f = 0$. With $\vec{D} = -\varepsilon\vec{\nabla}\phi$,

$$\vec{\nabla}\cdot\vec{D} = -\varepsilon\,\nabla^2\phi = 0,$$

so the potential satisfies Laplace's equation in both media,

$$\boxed{\nabla^2\phi_1 = 0, \qquad \nabla^2\phi_2 = 0.} \tag{3.4}$$

Normal component from the divergence theorem. Let $\hat{n}$ be the unit normal to the interface, pointing from medium 1 into medium 2, with $\partial/\partial n = \hat{n}\cdot\vec{\nabla}$ and $D_n = \vec{D}\cdot\hat{n}$. A pillbox of face area ΔS and height h straddles the interface. The divergence theorem, with the surface element $d\vec{S}$, gives

$$\oint_{\text{pillbox}} \vec{D}\cdot d\vec{S} = \int_{\text{pillbox}} \vec{\nabla}\cdot\vec{D}\; d^3r = 4\pi Q_f = 0.$$

As $h \to 0$, the side wall drops out and only the two faces remain,

$$(D_{2,n} - D_{1,n})\,\Delta S = 0 \quad \Rightarrow \quad \varepsilon_1 E_{1,n} = \varepsilon_2 E_{2,n}.$$

With $E_n = -\partial\phi/\partial n$,

$$\boxed{\varepsilon_1 \frac{\partial\phi_1}{\partial n} = \varepsilon_2 \frac{\partial\phi_2}{\partial n}.} \tag{3.5}$$

The same pillbox applied to $\vec{E}$ encloses the bound surface charge, $E_{2,n} - E_{1,n} = 4\pi\sigma_b$, so E_n itself jumps across the interface.

Tangential component from Stokes' theorem. In electrostatics, $\partial\vec{B}/\partial t = 0$, so Faraday's law gives $\vec{\nabla}\times\vec{E} = 0$. Let C be a rectangular loop with two sides of length Δs along a tangent $\hat{t}$ of

the interface, one in each medium, and two sides of height h crossing it. Stokes' theorem, with the line element $d\vec{s}$, gives

$$\oint_C \vec{E}\cdot d\vec{s} = \int_{S_C} \left(\vec{\nabla}\times\vec{E}\right)\cdot d\vec{S} = 0,$$

where S_C is the surface bounded by C. As $h \to 0$, only the two long sides remain,

$$(E_{2,t} - E_{1,t})\,\Delta s = 0 \quad \Rightarrow \quad E_{1,t} = E_{2,t}.$$

The same theorem forbids closed field lines, because $\vec{E}\cdot d\vec{s} > 0$ along a field line and $\oint \vec{E}\cdot d\vec{s}$ would be positive. Across the interface, the potential difference over the height h vanishes for a finite field,

$$\phi_2 - \phi_1 = -\int_0^h \vec{E}\cdot\hat{n}\,ds \;\longrightarrow\; 0 \qquad (h\to 0),$$

which gives

$$\boxed{\phi_1 = \phi_2 \quad \text{on the interface.}} \tag{3.6}$$

Since $E_t = -\hat{t}\cdot\vec{\nabla}\phi$ and $\phi_1 = \phi_2$ holds all along the interface, this condition contains $E_{1,t} = E_{2,t}$.

3.3 Dielectric Sphere in a Dielectric Medium

Setup of the boundary-value problem. Many physical systems can be modeled as a dielectric sphere of radius R_0 and dielectric constant ε_1 embedded in a dielectric medium of dielectric constant ε_2. A uniform external field $\vec{E}_0 = E_0\,\hat{z}$ is applied far from the sphere (Fig. 3). From now on, (r,θ,φ) are the spherical coordinates of the field point, measured from the center of the sphere with the polar axis along $\vec{E}_0$. The field point is therefore $\vec{r}$, which replaces $\vec{R}$ of Sections 1 and 2. The azimuthal angle φ is distinct from the potential ϕ. A rotation about the z axis leaves the setup unchanged, so the potential is azimuthally symmetric,

$$\phi = \phi(r,\theta), \qquad \frac{\partial\phi}{\partial\varphi} = 0.$$

The goal is to find ϕ, and therefore $\vec{E} = -\vec{\nabla}\phi$, in the whole space.

General azimuthally symmetric solution of Laplace's equation. With no free charge, ϕ satisfies Laplace's equation inside and outside the sphere. Its general azimuthally symmetric solution is

$$\boxed{\phi(r,\theta) = \sum_{l=0}^{\infty}\left(A_l\,r^l + \frac{B_l}{r^{l+1}}\right)\mathcal{P}_l(\cos\theta),} \tag{3.7}$$

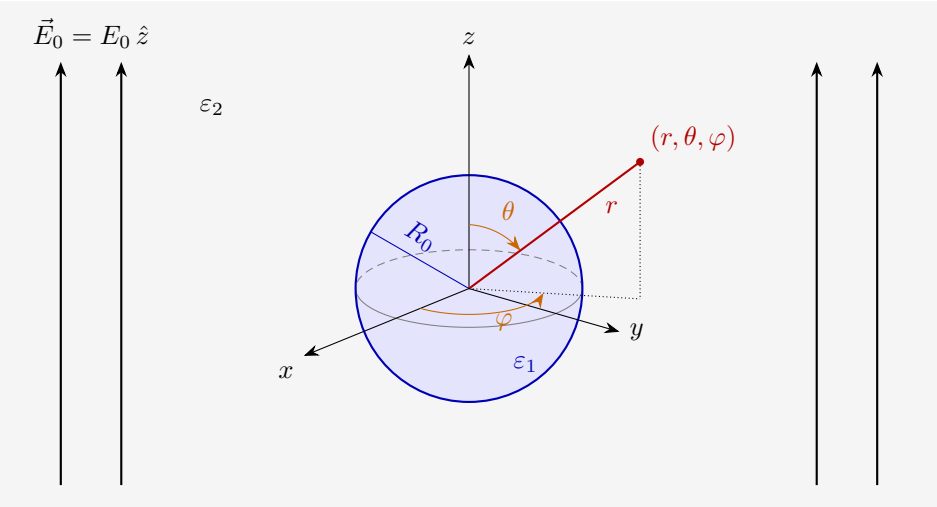


Figure 3: Dielectric sphere of radius R_0 and dielectric constant ε_1 in a medium of dielectric constant ε_2. The external field $\vec{E}_0 = E_0\hat{z}$ is uniform far from the sphere. A field point has the spherical coordinates (r, θ, φ), measured from the center of the sphere.

where $\mathcal{P}_l(x)$ is the Legendre polynomial of degree l, given by Rodrigues' formula

$$\mathcal{P}_l(x) = \frac{1}{2^l\, l!}\,\frac{d^l}{dx^l}\left(x^2 - 1\right)^l.$$

Separation of the radial and angular dependences. Each term of the sum is a product of a radial function and an angular function,

$$\phi_l(r, \theta) = \left(A_l\, r^l + \frac{B_l}{r^{l+1}}\right)\mathcal{P}_l(\cos\theta).$$

The angular profile of a single term is therefore the same at every radius. The full potential, however, mixes terms with different radial dependences, so its angular profile changes with r.

Orthogonality of the Legendre polynomials. The Legendre polynomials are orthogonal on $0 \le \theta \le \pi$ with the weight $\sin\theta$,

$$\int_0^\pi \mathcal{P}_l(\cos\theta)\,\mathcal{P}_m(\cos\theta)\,\sin\theta\,d\theta = \int_{-1}^{1}\mathcal{P}_l(x)\,\mathcal{P}_m(x)\,dx = \frac{2}{2l+1}\,\delta_{lm}.$$

The solution is therefore an expansion of the angular dependence in an orthogonal basis, in the same way as a Fourier series.

Electric field of the general solution. In spherical coordinates with azimuthal symmetry, the field is

$$\vec{E} = -\vec{\nabla}\phi = -\hat{r}\,\frac{\partial \phi}{\partial r} - \hat{\theta}\,\frac{1}{r}\frac{\partial \phi}{\partial \theta}.$$

With $d\mathcal{P}_l(\cos\theta)/d\theta = -\sin\theta\,\mathcal{P}_l'(\cos\theta)$, where the prime denotes the derivative with respect to the argument, the general solution gives

$$\boxed{\begin{aligned} E_r(r,\theta) &= \sum_{l=0}^{\infty}\left[-l\,A_l\,r^{l-1} + (l+1)\frac{B_l}{r^{l+2}}\right]\mathcal{P}_l(\cos\theta), \\ E_\theta(r,\theta) &= \sum_{l=0}^{\infty}\left(A_l\,r^{l-1} + \frac{B_l}{r^{l+2}}\right)\sin\theta\,\mathcal{P}_l'(\cos\theta). \end{aligned}} \tag{3.8}$$

Separate expansions inside and outside the sphere. The potential is continuous at $r = R_0$, but its radial derivative jumps there when $\varepsilon_1 \neq \varepsilon_2$. A single expansion therefore cannot hold in the whole space, and each region carries its own coefficients. From now on, $A_l^{(1)}, B_l^{(1)}$ are the coefficients of ϕ_1 for $r < R_0$, and $A_l^{(2)}, B_l^{(2)}$ are those of ϕ_2 for $r > R_0$. The corresponding fields are $\vec{E}_1 = -\vec{\nabla}\phi_1$ and $\vec{E}_2 = -\vec{\nabla}\phi_2$.

Finite potential at the center. The term $B_l^{(1)}/r^{l+1}$ diverges as $r \to 0$, but the center of the sphere contains no charge. Hence

$$B_l^{(1)} = 0 \quad \text{for all } l, \qquad \phi_1(r,\theta) = \sum_{l=0}^{\infty} A_l^{(1)}\,r^l\,\mathcal{P}_l(\cos\theta).$$

Uniform field far from the sphere. Far from the sphere, the field is the unperturbed external field, $\phi_2 \to -E_0\,r\cos\theta = -E_0\,r\,\mathcal{P}_1(\cos\theta)$. The terms $B_l^{(2)}/r^{l+1}$ vanish as $r \to \infty$, whereas $A_l^{(2)} r^l$ must reproduce this limit. Hence

$$A_1^{(2)} = -E_0, \qquad A_l^{(2)} = 0 \quad (l \geq 2),$$

and $A_0^{(2)} = 0$ by the choice of the potential reference. The coefficients $B_l^{(2)}$ remain free,

$$\phi_2(r,\theta) = -E_0\,r\cos\theta + \sum_{l=0}^{\infty}\frac{B_l^{(2)}}{r^{l+1}}\,\mathcal{P}_l(\cos\theta).$$

Matching conditions at the surface of the sphere. The surface $r = R_0$ has the normal $\hat{n} = \hat{r}$. Continuity of the potential, which contains the tangential condition $E_{1,\theta} = E_{2,\theta}$, and continuity of

the normal displacement, $\varepsilon_1\,\partial\phi_1/\partial r = \varepsilon_2\,\partial\phi_2/\partial r$, give for every θ

$$\boxed{\begin{aligned}\sum_{l=0}^{\infty} A_l^{(1)} R_0^l\,\mathcal{P}_l(\cos\theta) &= -E_0 R_0\cos\theta + \sum_{l=0}^{\infty}\frac{B_l^{(2)}}{R_0^{l+1}}\,\mathcal{P}_l(\cos\theta),\\ \varepsilon_1\sum_{l=0}^{\infty} l\,A_l^{(1)} R_0^{l-1}\,\mathcal{P}_l(\cos\theta) &= \varepsilon_2\left[-E_0\cos\theta - \sum_{l=0}^{\infty}(l+1)\frac{B_l^{(2)}}{R_0^{l+2}}\,\mathcal{P}_l(\cos\theta)\right].\end{aligned}} \tag{3.9}$$

Matching the coefficients of each Legendre polynomial. With $-E_0R_0\cos\theta = -E_0R_0\,\mathcal{P}_1(\cos\theta)$, moving every term of the two matching conditions to the left side gives

$$\sum_{l=0}^{\infty} c_l^{(\phi)}\,\mathcal{P}_l(\cos\theta) = 0, \qquad \sum_{l=0}^{\infty} c_l^{(D)}\,\mathcal{P}_l(\cos\theta) = 0 \qquad \text{for all } \theta,$$

where each coefficient is the difference between the two sides,

$$\begin{aligned} c_l^{(\phi)} &= A_l^{(1)} R_0^l - \frac{B_l^{(2)}}{R_0^{l+1}} + E_0 R_0\,\delta_{l1},\\ c_l^{(D)} &= \varepsilon_1\,l\,A_l^{(1)} R_0^{l-1} + \varepsilon_2\,(l+1)\,\frac{B_l^{(2)}}{R_0^{l+2}} + \varepsilon_2 E_0\,\delta_{l1}.\end{aligned}$$

The Kronecker delta δ_{l1} places the external-field term in the $l = 1$ coefficient only. Multiplying either sum by $\mathcal{P}_m(\cos\theta)\sin\theta$ and integrating over $0 \le \theta \le \pi$, the orthogonality relation leaves only the term $l = m$,

$$\sum_{l=0}^{\infty} c_l \int_0^{\pi} \mathcal{P}_l(\cos\theta)\,\mathcal{P}_m(\cos\theta)\sin\theta\,d\theta = \frac{2}{2m+1}\,c_m = 0, \qquad c_l = c_l^{(\phi)} \text{ or } c_l^{(D)}.$$

Hence $c_l^{(\phi)} = 0$ and $c_l^{(D)} = 0$ for every l. For $l \neq 1$,

$$\begin{aligned} A_l^{(1)} R_0^l = \frac{B_l^{(2)}}{R_0^{l+1}} \quad &\Rightarrow \quad A_l^{(1)} = \frac{B_l^{(2)}}{R_0^{2l+1}},\\ \varepsilon_1\,l\,A_l^{(1)} R_0^{l-1} &= -\varepsilon_2\,(l+1)\,\frac{B_l^{(2)}}{R_0^{l+2}}.\end{aligned}$$

For $l = 1$, the external field adds a term to each condition,

$$\begin{aligned} A_1^{(1)} R_0 &= -E_0 R_0 + \frac{B_1^{(2)}}{R_0^2},\\ \varepsilon_1 A_1^{(1)} &= -\varepsilon_2 E_0 - 2\,\varepsilon_2\,\frac{B_1^{(2)}}{R_0^3}.\end{aligned}$$

Solution of the matching conditions. For $l \neq 1$, substituting $A_l^{(1)} = B_l^{(2)}/R_0^{2l+1}$ into the displacement condition gives

$$[\varepsilon_1\, l + \varepsilon_2\,(l+1)]\,\frac{B_l^{(2)}}{R_0^{l+2}} = 0.$$

The bracket is positive for $\varepsilon_1, \varepsilon_2 > 0$, so $B_l^{(2)} = 0$ and therefore $A_l^{(1)} = 0$. For $l = 1$, the potential condition gives $A_1^{(1)} = -E_0 + B_1^{(2)}/R_0^3$. Substituting it into the displacement condition gives

$$(\varepsilon_1 + 2\varepsilon_2)\,\frac{B_1^{(2)}}{R_0^3} = (\varepsilon_1 - \varepsilon_2)\,E_0.$$

The coefficients are therefore

$$\boxed{A_l^{(1)} = B_l^{(2)} = 0 \quad (l \neq 1), \qquad A_1^{(1)} = -\frac{3\varepsilon_2}{\varepsilon_1 + 2\varepsilon_2}\,E_0, \qquad B_1^{(2)} = \frac{\varepsilon_1 - \varepsilon_2}{\varepsilon_1 + 2\varepsilon_2}\,E_0 R_0^3.} \tag{3.10}$$

Potential inside and outside the sphere. Substituting the coefficients into ϕ_1 and ϕ_2 gives the potential in the whole space,

$$\boxed{\phi(r,\theta) = \begin{cases} -\dfrac{3\varepsilon_2}{\varepsilon_1 + 2\varepsilon_2}\,E_0\, r\cos\theta, & r < R_0, \\[2ex] \left(\dfrac{\varepsilon_1 - \varepsilon_2}{\varepsilon_1 + 2\varepsilon_2}\,\dfrac{R_0^3}{r^2} - r\right) E_0 \cos\theta, & r \geq R_0. \end{cases}} \tag{3.11}$$

Uniform field inside the sphere. Inside, $r\cos\theta = z$, so the potential is linear in z and the field is

$$\boxed{\vec{E}_1 = -\vec{\nabla}\phi_1 = \frac{3\varepsilon_2}{\varepsilon_1 + 2\varepsilon_2}\,\vec{E}_0.} \tag{3.12}$$

The field inside the sphere is uniform and parallel to $\vec{E}_0$. For $\varepsilon_1 > \varepsilon_2$, the field inside is weaker than E_0.

Outside field as the external field plus a dipole field. Outside, the potential splits into the potential of the external field and a correction,

$$\phi_2 = -E_0\, r\cos\theta + \frac{\varepsilon_1 - \varepsilon_2}{\varepsilon_1 + 2\varepsilon_2}\,E_0 R_0^3\,\frac{\cos\theta}{r^2} = -\vec{E}_0 \cdot \vec{r} + \frac{\vec{p}_{\text{eff}} \cdot \hat{r}}{r^2},$$

where $\hat{z} \cdot \hat{r} = \cos\theta$ and

$$\vec{p}_{\text{eff}} = \frac{\varepsilon_1 - \varepsilon_2}{\varepsilon_1 + 2\varepsilon_2}\,R_0^3\,\vec{E}_0.$$

The correction has the form of the dipole potential $\hat{R}\cdot\vec{p}/R^2$ of Section 1, with $\vec{R}$ replaced by $\vec{r}$. Its field follows from $E_r = -\partial\phi/\partial r$ and $E_\theta = -(1/r)\,\partial\phi/\partial\theta$,

$$\vec{E}_2 = E_0\cos\theta\left(1 + 2\,\frac{\varepsilon_1-\varepsilon_2}{\varepsilon_1+2\varepsilon_2}\,\frac{R_0^3}{r^3}\right)\hat{r} - E_0\sin\theta\left(1 - \frac{\varepsilon_1-\varepsilon_2}{\varepsilon_1+2\varepsilon_2}\,\frac{R_0^3}{r^3}\right)\hat{\theta}.$$

The terms without R_0 give $E_0(\cos\theta\,\hat{r} - \sin\theta\,\hat{\theta}) = E_0\hat{z} = \vec{E}_0$. With $\vec{p}_{\rm eff} = p_{\rm eff}\,\hat{z}$, the remaining terms are

$$\frac{p_{\rm eff}}{r^3}\left(2\cos\theta\,\hat{r} + \sin\theta\,\hat{\theta}\right).$$

With $\sin\theta\,\hat{\theta} = \cos\theta\,\hat{r} - \hat{z}$ and $\cos\theta = \hat{z}\cdot\hat{r}$, the bracket becomes

$$2\cos\theta\,\hat{r} + \sin\theta\,\hat{\theta} = 2(\hat{z}\cdot\hat{r})\,\hat{r} + (\hat{z}\cdot\hat{r})\,\hat{r} - \hat{z} = 3(\hat{z}\cdot\hat{r})\,\hat{r} - \hat{z}.$$

Multiplying by $p_{\rm eff}$ turns $\hat{z}$ into $\vec{p}_{\rm eff}$, so the outside field is

$$\boxed{\vec{E}_2 = \vec{E}_0 + \frac{3(\vec{p}_{\rm eff}\cdot\hat{r})\,\hat{r} - \vec{p}_{\rm eff}}{r^3}, \qquad \vec{p}_{\rm eff} = \frac{\varepsilon_1-\varepsilon_2}{\varepsilon_1+2\varepsilon_2}\,R_0^3\,\vec{E}_0,} \tag{3.13}$$

which is the external field plus the field of a point dipole $\vec{p}_{\rm eff}$ at the center of the sphere.

Consistency checks of the solution. At $r = R_0$, the tangential field and the normal displacement are continuous,

$$\begin{aligned} E_{2,\theta}(R_0) &= -E_0\sin\theta\left(1 - \frac{\varepsilon_1-\varepsilon_2}{\varepsilon_1+2\varepsilon_2}\right) = -\frac{3\varepsilon_2}{\varepsilon_1+2\varepsilon_2}\,E_0\sin\theta = E_{1,\theta}(R_0), \\ \varepsilon_2\,E_{2,r}(R_0) &= \varepsilon_2\,E_0\cos\theta\left(1 + 2\,\frac{\varepsilon_1-\varepsilon_2}{\varepsilon_1+2\varepsilon_2}\right) = \frac{3\varepsilon_1\varepsilon_2}{\varepsilon_1+2\varepsilon_2}\,E_0\cos\theta = \varepsilon_1\,E_{1,r}(R_0). \end{aligned}$$

Far from the sphere, the dipole term decays as r^{-3} and $\vec{E} \to \vec{E}_0$. Three limits reproduce known results,

$$\begin{aligned} \varepsilon_1 = \varepsilon_2: &\qquad \vec{E} = \vec{E}_0 \text{ everywhere}, \qquad \vec{p}_{\rm eff} = 0, \\ \varepsilon_1 \to \infty: &\qquad \vec{E}_1 \to 0, \qquad \phi(R_0,\theta) = 0, \qquad \vec{p}_{\rm eff} \to R_0^3\,\vec{E}_0, \\ \varepsilon_1 = 1,\ \varepsilon_2 > 1: &\qquad \vec{E}_1 = \frac{3\varepsilon_2}{1+2\varepsilon_2}\,\vec{E}_0, \qquad |\vec{E}_1| > E_0. \end{aligned}$$

The first limit describes a sphere that is indistinguishable from its surroundings. The second describes a conducting sphere, which screens its interior and forms an equipotential surface. The third describes a spherical cavity in a dielectric, where the field is stronger than the external field.

Effective dipole moment and the dipole moment of the sphere. The interface carries bound charge from both media. With the outward normal $\hat{r}$ of the sphere, medium 1 contributes $\vec{P}_1 \cdot \hat{r}$ and medium 2 contributes $-\vec{P}_2 \cdot \hat{r}$. With $\vec{P} = (\varepsilon - 1)\vec{E}/4\pi$ and $\varepsilon_1 E_{1,r} = \varepsilon_2 E_{2,r}$, the total bound surface charge is

$$\sigma_{b,\mathrm{tot}} = \left(\vec{P}_1 - \vec{P}_2\right) \cdot \hat{r} = \frac{E_{2,r} - E_{1,r}}{4\pi} = \frac{3}{4\pi} \frac{\varepsilon_1 - \varepsilon_2}{\varepsilon_1 + 2\varepsilon_2} E_0 \cos\theta.$$

A surface charge $\sigma_0 \cos\theta$ on a sphere has the dipole moment $\oint \sigma_0 \cos\theta \, \vec{r} \, dS = \frac{4\pi}{3} R_0^3 \, \sigma_0 \, \hat{z}$, so $\sigma_{b,\mathrm{tot}}$ produces exactly $\vec{p}_{\mathrm{eff}}$. The sphere's own material, however, carries only $\vec{P}_1$, and its dipole moment is

$$\boxed{\vec{p}_{\mathrm{sphere}} = \frac{4\pi}{3} R_0^3 \, \vec{P}_1 = \frac{(\varepsilon_1 - 1)\, \varepsilon_2}{\varepsilon_1 + 2\varepsilon_2} \, R_0^3 \, \vec{E}_0 \;\neq\; \vec{p}_{\mathrm{eff}} \quad (\varepsilon_2 \neq 1).} \tag{3.14}$$

The difference is the dipole moment of the bound charge of medium 2 at the interface. The two moments coincide only for a sphere in vacuum, $\varepsilon_2 = 1$. The field therefore depends only on the ratio $\varepsilon_1/\varepsilon_2$, whereas the polarization of the sphere does not.